

Stage 1 — The problem, the pilot and the protocol

Where the Shortcut Sits: progress report 1

In one sentence. Region-of-interest (ROI) masking is the standard defence against acquisition artifacts that act as shortcuts, but its rationale assumes that the artifact lies outside the ROI; the pilot study suggested that masking turns from helpful to harmful when the artifact overlaps the lesion. This report re-implements the pilot from its written specification, compares every reported number (76 of 87 match, 5 do not), fixes the experimental protocol, and replaces the pilot’s interval estimator, which was too optimistic, by a crossed seed×image bootstrap.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma.
$A \in \{0, 1\}$	artifact present: a ruler scale bar (synthetic, pasted) or hair (real).
ROI	region of interest: the skin lesion (manual or U-Net lesion mask); drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{outside}} - [\text{mask} - \text{ERM}]_{\text{inside}}$ of reversed-test AUROC: in the ruler sweep the <i>location interaction</i> $[\cdot]_{r=0} - [\cdot]_{r=1}$; in the real hair traps Trap B – Trap A.
CI	95% crossed seed×image bootstrap interval (Sec. 5.2); every interval in this report uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.

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1 Purpose of this report

This report covers the first block of work: understanding the clinical problem, re-deriving the pilot study (experiments E1–E15) from scratch, and fixing a protocol that is safe against the two most common ways a shortcut study fools itself — data leakage between training and test images, and too-narrow uncertainty intervals. The pilot’s claims are treated as hypotheses: each reported number was re-computed by new code written from the pilot’s specification, and the two sets of numbers were compared one by one. Nothing in this report depends on results that were not re-derived.

2 The clinical problem

2.1 Shortcut artifacts

Deep classifiers exploit any feature that predicts the label in the training data, including features that are not evidence of disease (Geirhos et al., 2020; DeGrave et al., 2021). In dermoscopy, surgical skin markings (Winkler et al., 2019) and ruler scale bars (Winkler et al., 2021) changed the melanoma predictions of a market-approved system; hair and dark corners are learned as well (Wang et al., 2024; Pewton et al., 2024). In chest radiography, pneumothorax classifiers detect the chest drain that treats the condition (Oakden-Rayner et al., 2020; Jiménez-Sánchez et al., 2023), and a pneumonia model learned the hospital that took the image (Zech et al., 2018). Such associations are produced by the clinical workflow itself: suspicious lesions are measured and marked, so the measuring device appears more often on malignant cases. They are therefore present in every retrospective dataset, and a model that relies on them fails exactly on the patients whose workflow differs — for example a melanoma that was photographed without a ruler (Brown et al., 2023).

2.2 The standard defence and its hidden assumption

The most common defence is to restrict the classifier to the ROI: segment the lesion and discard the background (“ROI masking”). It is attractive because it needs only a lesion mask, which many pipelines compute anyway, and no knowledge of the artifact. Its rationale is that the shortcut lives in the background. Rulers, hair and ink markings do not respect that assumption: they are placed on, across or next to the lesion. When an artifact overlaps the ROI, masking cannot remove it — and it removes everything *else* that competes with it.

2.3 Research question

Does the benefit of ROI masking depend on where the artifact sits relative to the ROI — and if so, by how much, through which mechanism, and does it matter for real artifacts?

For this stage the question is narrower: **does the pilot’s evidence for a location-dependent effect of masking survive an independent re-implementation, and is the protocol that produced it sound?**

3 Why the study is designed this way

- **Control the overlap first.** The pilot pasted a synthetic ruler at a controlled overlap with the lesion (0–100%, Figure 1). This isolates location from everything else: the same images, the same ruler, only its position changes. Starting with real artifacts only was rejected, because the images that carry an artifact on the lesion differ from those that carry it beside the lesion, so a location effect could not be separated from a population difference.

- **Frozen foundation-model features and linear heads.** This is how foundation models are typically used on small medical datasets; it makes every arm comparable (same features, only the head or the input changes); and it is cheap enough to repeat over seeds, overlaps and arms. End-to-end fine-tuning multiplies the cost by about two orders of magnitude and was not part of the pilot.
- **A trap with a reversed test.** Training with a strong, known artifact–label association (0.9/0.1) and testing on the reversed association (0.1/0.9) measures reliance on the artifact directly: a model that uses it must fall on the reversed test (Figure 3). The correlated and clean tests measure what a mitigation costs. A single natural test set was rejected as the main design because its shortcut is too weak to measure a location effect precisely.
- **AUROC as the primary metric.** It does not depend on a decision threshold, so a mitigation cannot look good merely by moving the threshold. Thresholds are still fixed (on clean validation data) for secondary metrics.
- **An artifact-free comparison group for real artifacts (E13).** The pilot’s first real-hair design (E12) compared in-lesion hair directly with out-of-lesion hair; E13 instead builds two traps that share one hair-free group. Section 7.4 shows why the difference matters.

4 Data

Controlled cohort (ISIC 2018) ISIC 2018 Task 1/2 images with manual lesion masks (Codella et al., 2019); melanoma labels from the union of the ISIC 2016 and 2017 ground truth (2017 takes precedence; three images with conflicting labels dropped). A 65×19 px ruler (at 518 px) must be placeable at 0, 25, 50, 75 and 100% overlap with the lesion within 10 percentage points; 2,437 images qualify. The excluded images have very small lesions (a ruler cannot lie fully inside them) and more melanomas; their placements are frozen in `frozen/placements_518.65x19.json` and reused by every run.

Real-hair cohort (ISIC 2019) 25,331 dermoscopic images from three hospitals (HAM10000 (Tschandl et al., 2018), BCN20000 (Combalia et al., 2019), MSK), melanoma versus all other diagnoses. Hair masks: the published semi-automatic hair/ruler masks of Wegley et al. (2026). Lesion masks: HAM10000 manual masks where available, otherwise a U-Net trained only on ISIC 2018 Task 1 (Table 6). Leakage groups: lesion identifier $\cup$ perceptual hash.

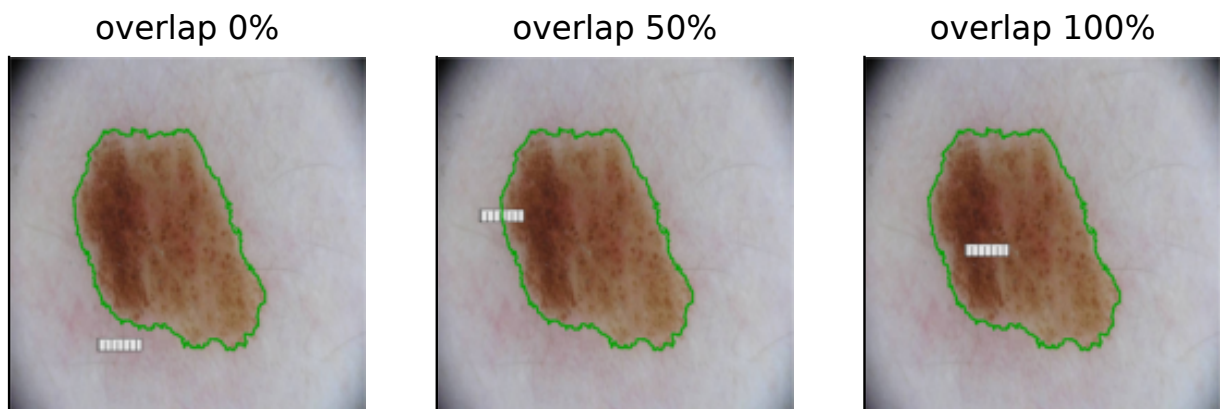


Figure 1: The controlled sweep: the same ISIC 2018 image with the synthetic ruler at 0, 50 and 100% overlap with the lesion (green: manual lesion mask). ISIC images CC BY-NC 4.0.



Figure 2: The real hair traps (ISIC 2019): hair on the lesion (Trap A) and beside it (Trap B); the green line is the lesion outline. The hair masks are not drawn because the published masks carry no licence that allows redistribution. ISIC images CC BY-NC 4.0.

5 The protocol

Table 1 lists every component with the reason it was chosen; Figure 3 shows the four test environments.

Table 1: The protocol fixed in this stage.

Component	Choice	Why
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: no artifact (synthetic) or artifact independent of the label (real)	a strong, known shortcut whose use is exposed when the association is reversed
Primary metric	reversed-test AUROC; clean and correlated AUROC reported beside it	threshold-free; a model that uses the artifact must fall on the reversed test
Encoders	frozen DINOv2 ViT-B/14 at 518 px (CLS token), DINOv2 at 224 px, DermLIP (dermatology CLIP model) at 224 px	how foundation models are used on small medical data; every arm sees the same features
Heads	logistic regression (liblinear, class-balanced); $C \in \{0.01, 0.1, 1, 10\}$ and the decision threshold (maximal balanced accuracy) chosen on clean validation data only	no test or artifact information enters model selection
Arms	ERM; ROI masking; oracle inpainting of the artifact; group-balanced training; DFR; paired and unpaired LEACE; GroupDRO; prevalence calibration	masking is compared with methods that target the artifact itself
Splits	grouped by lesion identifier $\cup$ perceptual hash (Hamming ≤ 8); no group crosses a split	near-duplicates across a split inflate shortcut measurements (Sec. 5.1)
Seeds	3 seeds (42, 123, 456) for the synthetic sweeps; 5 seeds $\times$ 5 grouped folds for the real hair traps	training and split variation are both part of the uncertainty
Uncertainty	crossed seed $\times$ image bootstrap, 10,000 replicates, percentile 95% intervals	seeds share test images (Sec. 5.2)
Registration	design, claims and support criteria committed to the repository before each verdict-producing run	separates confirmation from exploration

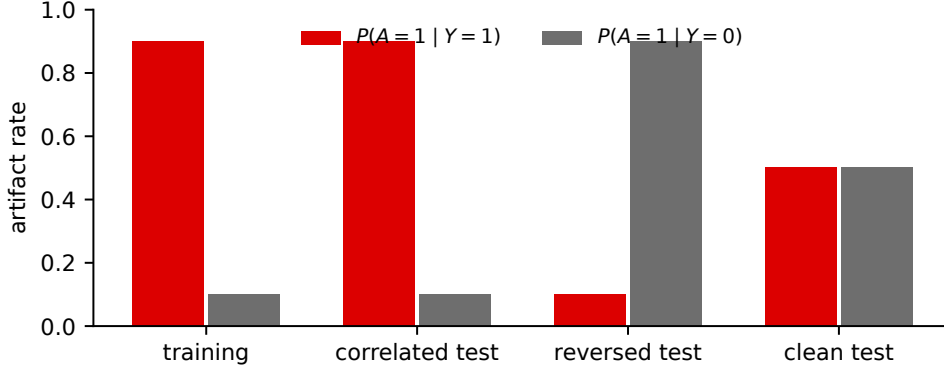


Figure 3: The trap environments. The training set and the correlated test share a strong artifact–label association; the reversed test inverts it; the clean test removes it. A model that uses the artifact gains on the correlated test and collapses on the reversed one.

Arms *ERM*: logistic head on the unmasked image. *ROI masking*: pixels outside the lesion mask are set to the image mean before feature extraction, at training and at test time. *Oracle inpainting*: the known artifact pixels are inpainted (Telea, 2004) — an upper bound that needs the artifact mask. *Group-balanced*: the four artifact×label cells are equally weighted. *DFR*: the last layer is re-fitted on a group-balanced subset of the validation fold (Kirichenko et al., 2023). *LEACE*: closed-form linear erasure of the artifact concept (Belrose et al., 2023), either *paired* (original versus artifact-removed versions of the same images) or *unpaired* (image-level artifact labels). *GroupDRO* (Sagawa et al., 2020). *Prevalence calibration* (Kina and Petersen, 2026). Balanced training, DFR, GroupDRO and unpaired LEACE need image-level artifact labels; inpainting and paired LEACE need artifact masks; masking needs only the lesion mask.

5.1 Leakage: grouped versus random splits

Near-duplicate photographs of one lesion are common in ISIC. If one copy is in training and another in test, the test measures memorisation. Every split therefore groups images by lesion identifier and perceptual hash (64-bit pHash, Hamming distance ≤ 8) and checks that no group crosses a split. To measure what this protects against, the controlled sweep was repeated with five random image-level splits, everything else fixed.

Result Random splits leave near-duplicate images on both sides of the split and inflate the measured ERM shortcut gap by 18% (no overlap) and 10% (full overlap), and shift both masking effects (Table 2); every analysis therefore uses grouped splits.

Table 2: Leakage: grouped versus random image-level splits (controlled sweep, DINOv2@518).

Overlap	Split	Near-duplicate pairs across split	ERM gap (corr – rev)	mask – ERM (rev)
0%	grouped	0	0.241	+0.184
	random (mean of 5)	93	0.284	+0.223
100%	grouped	0	0.421	−0.129
	random (mean of 5)	93	0.465	−0.088

5.2 Uncertainty: the corrected (crossed) bootstrap

Every comparison is a paired difference in AUROC between two arms on the same test images, averaged over training seeds (or cross-validation folds). The pilot’s estimator resampled the seeds and then, *independently within each sampled seed*, the test images. In these designs, however, every

seed scores the *same* test images: the controlled sweep uses one fixed test split, and in the hair traps the pooled cross-validation folds of every seed cover the same image pool. Resampling the images independently per seed then treats one test set as several and divides the test-sampling variance by the number of seeds, so the intervals are too narrow.

Why it matters With three seeds and one test set, the per-seed estimator behaves as if the test set were three times larger than it is. The problem is not specific to this project: any study that averages several training seeds on one test set and resamples the images per seed has it.

The crossed estimator Each replicate resamples the seeds with replacement and draws one $\text{Poisson}(1)$ weight per test *image*, shared by every seed (and both arms) in which that image appears; the per-seed statistic is recomputed with those weights and averaged (`wtss.stats_crossed`; 10,000 replicates; percentile intervals). The image weights make the test-set variance count once, as it should; the seed resampling keeps the training variance. The estimator was registered before any interval was recomputed (85da735, 2026-09-27 23:43; applied everywhere under c45d482, 2026-09-28 00:35).

Consequences for the pilot On the 178 pilot intervals that both estimators can compute, the crossed interval is wider by a median factor of 1.45 (interquartile range 1.32–1.61; Figure 4, Table 3). 7 intervals that excluded zero with the pilot’s estimator no longer do, and 1 now excludes zero (Table 4). Point estimates are identical: only the uncertainty changed.

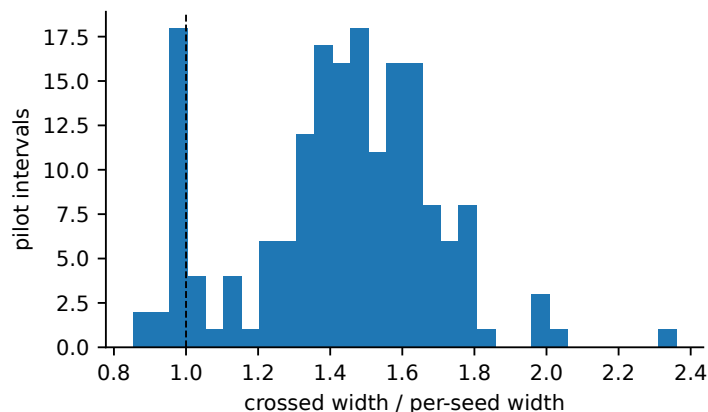


Figure 4: Width of the crossed interval divided by the width of the pilot’s per-seed interval, for every pilot contrast (same predictions).

Table 3: The pilot’s analyses under both interval estimators (same predictions, same point estimates; 10,000 replicates each). Width ratio > 1: the crossed interval is wider.

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ratio (crossed / per-seed)
analysis/E14_area_ratio	4	0		0		1.63
spec_e13/dermlip224_spec	29	1		0		1.45
spec_e13/dino518_e12_contrast	3	0		0		1.44
spec_e13/dino518_spec	28	0		1		1.54
synthetic/isc2018	114	6		0		1.44
all	178	7		1		1.45

Table 4: The pilot contrasts whose verdict (interval excludes 0 or not) depends on the estimator. Every other contrast keeps its verdict.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
synthetic/isic2018/ dino518_ruler_fixed_corr_main	0.25 / mask / erm / test_rev	−0.059 [−0.127, +0.008]	no longer excludes 0
synthetic/isic2018/ dino518_ruler_fixed_occlusion_occlusion	0.0 / mask / erm / test_uncorr	+0.036 [−0.016, +0.086]	no longer excludes 0
synthetic/isic2018/ dino518_ruler_fixed_occlusion_occlusion	0.5 / mask / erm / test_uncorr	+0.041 [−0.010, +0.090]	no longer excludes 0
synthetic/isic2018/ dino518_ruler_fixed_occlusion_occlusion	0.75 / mask / erm / test_uncorr	+0.037 [−0.013, +0.086]	no longer excludes 0
synthetic/isic2018/ dino518_ruler_fixed_occlusion_occlusion	1.0 / mask / erm / test_uncorr	+0.039 [−0.010, +0.091]	no longer excludes 0
synthetic/isic2018/ dino224_ruler_fixed_corr_main	0.25 / mask / erm / test_rev	−0.049 [−0.117, +0.018]	no longer excludes 0
spec_e13/dino518_spec	trapA / mask / clean / all / mask / erm	−0.023 [−0.045, −0.001]	now excludes 0
spec_e13/dermlip224_spec	trapB / leace_unpaired / clean / all / leace_unpaired / erm	−0.019 [−0.046, +0.007]	no longer excludes 0

5.3 Pre-registration

Before every verdict-producing experiment, a document stating the design, the claims and what counts as support is committed to the repository; results are committed afterwards, and a claim is reported whichever way it goes. The pilot itself preceded this rule; its written specification (`docs/REPLICATION_SPEC.md`, first commit 2699d51, 2026-09-26 15:47) serves as its registration for the replication. **Caveat:** commit times are set by the machine that makes the commit and are not independent proof of order; registrations that are also pushed to GitHub before running carry GitHub’s time stamp as an external reference.

6 The pilot and its re-implementation

6.1 What the pilot tested

post hoc (the pilot preceded pre-registration; its specification was written down afterwards and re-implemented). Table 5 lists the fifteen experiments. The pilot made five claims: (i) masking removes an artifact that lies outside the lesion and makes the model worse than doing nothing when it lies inside; (ii) the harm is governed by how much of the artifact survives the mask (retention), not by occlusion or lost context; (iii) methods that need only an image-level artifact label (balanced training, DFR) fix both cases; (iv) linear erasure is insufficient for real artifacts under the sound (paired) protocol; (v) all of this holds for real hair with pre-registered criteria on two encoders.

Table 5: The pilot experiments (specification: `docs/REPLICATION_SPEC.md`).

Exp.	What it tests
E1	controlled ruler sweep, DINOv2@518: mask – ERM at 0–100% overlap; location interaction
E2	the same at 224px (DINOv2) and with the dermatology model DermLIP
E3	appearance randomisation (colour, opacity, ticks, scale of the ruler)
E4	occlusion control: ruler present but uncorrelated with the label (50/50)
E5	mask dilation: restoring ruler pixels around the lesion (retention)
E6	same-head counterfactual sensitivity $ \Delta p $ to inserting the ruler
E7	lesion-size strata (artifact-to-lesion area ratio)
E8	oracle inpainting and a prediction-consistency repair
E9	bridge: group-balanced training, DFR, GroupDRO, paired LEACE at 100% overlap
E10	hair decodability from the embedding; paired LEACE on real hair
E11	atlas of real artifacts (hair, ink, vignetting) in ISIC 2019 / HAM10000; vignetting trap
E12	overlap-contrast trap: in-lesion versus out-of-lesion hair, no artifact-free group
E13	real hair traps (Trap A in-lesion, Trap B out-of-lesion, shared hair-free group), ISIC 2019
E14	area-ratio strata of the real hair trap
E15	E13 with DermLIP

6.2 How it was re-implemented

registered before results (specification `docs/REPLICATION_SPEC.md`, 2699d51, 2026-09-26 15:47). The pilot’s code was not available for most experiments (E9–E15), so every experiment was re-implemented from the written specification in a new code base (`src/wtss`). A number *matches* (MATCH) if it has the same sign, the same verdict on whether its interval excludes zero, and differs by at most 0.02; SIGN+CI if sign and verdict agree but the difference is larger; MISMATCH otherwise. Where the specification was silent or impossible to follow, the choice made is recorded with its reason (Table 6). Two choices matter most: the hair-free group was defined as at most 30 hair pixels in the native mask — the only threshold that reproduces the pilot’s outcome-free cell counts — and, on the full ISIC 2019 cohort, pHash distance ≤ 8 chains unrelated lesions into one giant component, so the real-hair traps apply it within each trap pool, as the pilot’s counts imply.

Table 6: Where the re-implementation deviates from the pilot’s written specification, and why.

#	Specification	Re-implementation	Reason and effect
1	GPU V100 32 GB	RTX 3090 24 GB	Frozen features; per-seed AUROCs of the rerun match the archive within 0.0005 (E1, E8, E9).
2	Feature extraction per (seed, env)	per *view* (image $\times$ method $\times$ overlap), environments assembled by indexing	Mathematically identical (presence is a function of image/label/seed/env only); $\sim 10\times$ fewer backbone passes.
3	pHash ≤ 8 on the full ISIC 2019 cohort (repo-default analysis)	spec E13 run: pHash ≤ 8 within each matched trap pool (as the spec’s counts imply); exploratory runs: lesion_id $\cup$ pHash ≤ 2 on all 25,331	On all 25,331 images pHash ≤ 8 chains unrelated lesions into one component of 10,974 images (same-lesion precision at distance 8: 2%).
4	Hair-free A = 0 group (threshold not stated in spec)	≤ 30 hair/ruler pixels in the native-resolution archive mask	Calibrated on outcome-free counts only: reproduces A0.Y0 709 (spec 708), A0.Y1 157 (157) and all six source proportions to 3 decimals.
5	Archive mask file names	matched on zero-padded numeric id	Archive names are irregular (<code>_downsampled_mvig</code> , <code>_downsam_inkmark</code> , typos, truncated ids). 24 truncated ink ids flag their 10 candidate images as ink-uncertain (exploratory runs only).
6	Images without HAM/BCN lesion_id	source = MSK	Spec lists three sources; ISIC 2019’s non-HAM/BCN images are MSK (incl. <code>_downsampled</code>).
7	U-Net (Ronneberger 64–1024, Task 1 only)	same architecture (+BatchNorm), 256 px, 40 epochs, Adam 1e-4, BCE+Dice, 20% held-out pHash groups	Training hyper-parameters are not in the spec.
8	Derm1M from archive	fresh clone of the official repo	Archived copy lacks <code>bpe_simple_vocab_16e6.txt.gz</code> (.gz files were not archived); DermLIP cannot load without it.
9	§3.1 ”leakage inflated the gap by 27%”	not reproduced from the archive	Pre-correction pilot used the identical split; matched-resolution gap ratio 0.96–0.97. A controlled leakage experiment is run instead (<code>scripts/run_leakage_experiment.py</code>).

7 Results

7.1 Replication overview

Of the 87 numbers the pilot reported, 76 MATCH, 6 agree in sign and significance but differ by more than 0.02, and 5 MISMATCH (Table 7). Before the interval estimator was corrected the ledger read 73 / 12 / 3 of 88; the difference is entirely due to the wider intervals (Section 7.6).

Table 7: Replication ledger: every number the pilot reported, its re-derived value with the crossed 95% interval, and the verdict. The pilot’s own intervals came from the per-seed estimator (Sec. 5.2) and are not reproduced; only its point values are shown.

Exp.	Quantity	Pilot (point)	Replication (crossed CI)	Verdict
E1	mask – ERM rev @0%	+0.184	+0.184 [+0.117, +0.250]	MATCH
E1	mask – ERM rev @25%	-0.058	-0.059 [-0.127, +0.008]	MISMATCH
E1	mask – ERM rev @50%	-0.097	-0.097 [-0.164, -0.030]	MATCH
E1	mask – ERM rev @75%	-0.111	-0.111 [-0.181, -0.043]	MATCH
E1	mask – ERM rev @100%	-0.128	-0.129 [-0.195, -0.064]	MATCH
E1	interaction 0% – 100% (mask)	+0.312	+0.313 [+0.258, +0.370]	MATCH
E2	dino224 mask – ERM @0%	+0.364	+0.361 [+0.275, +0.444]	MATCH
E2	dino224 mask – ERM @25%	-0.040	-0.049 [-0.117, +0.018]	MATCH
E2	dino224 mask – ERM @50%	-0.076	-0.081 [-0.143, -0.019]	MATCH
E2	dino224 mask – ERM @75%	-0.102	-0.106 [-0.166, -0.044]	MATCH
E2	dino224 mask – ERM @100%	-0.092	-0.095 [-0.156, -0.035]	MATCH
E2	dermlip224 mask – ERM @0%	+0.160	+0.161 [+0.102, +0.222]	MATCH
E2	dermlip224 mask – ERM @25%	-0.291	-0.289 [-0.366, -0.215]	MATCH
E2	dermlip224 mask – ERM @50%	-0.300	-0.298 [-0.371, -0.225]	MATCH
E2	dermlip224 mask – ERM @75%	-0.313	-0.309 [-0.378, -0.239]	MATCH
E2	dermlip224 mask – ERM @100%	-0.268	-0.263 [-0.348, -0.180]	MATCH
E3	variable ruler mask – ERM @0%	+0.147	+0.146 [+0.082, +0.211]	MATCH
E3	variable ruler mask – ERM @50%	-0.094	-0.094 [-0.168, -0.021]	MATCH
E3	variable ruler mask – ERM @100%	-0.155	-0.155 [-0.228, -0.084]	MATCH
E3	variable ruler balanced – ERM @0%	+0.103	+0.102 [+0.068, +0.140]	MATCH
E3	variable ruler balanced – ERM @50%	+0.171	+0.171 [+0.132, +0.213]	MATCH

Exp.	Quantity	Pilot (point)	Replication (crossed CI)	Verdict
E3	variable ruler balanced – ERM @100%	+0.222	+0.222 [+0.181, +0.265]	MATCH
E4	occlusion (50/50) mask – ERM @0%	+0.035	+0.036 [-0.016, +0.086]	MISMATCH
E4	occlusion (50/50) mask – ERM @50%	+0.033	+0.041 [-0.010, +0.090]	MATCH
E4	occlusion (50/50) mask – ERM @100%	+0.039	+0.039 [-0.010, +0.091]	MISMATCH
E5	dilate0 – ERM @50%	-0.097	-0.097 [-0.164, -0.031]	MATCH
E5	dilate10 – ERM @50%	-0.138	-0.138 [-0.203, -0.074]	MATCH
E5	dilate25 – ERM @50%	-0.153	-0.153 [-0.221, -0.089]	MATCH
E5	dilate50 – ERM @50%	-0.142	-0.142 [-0.206, -0.083]	MATCH
E5	dilate0 – ERM @100%	-0.128	-0.129 [-0.195, -0.066]	MATCH
E5	dilate10 – ERM @100%	-0.143	-0.144 [-0.206, -0.082]	MATCH
E5	dilate25 – ERM @100%	-0.144	-0.144 [-0.209, -0.082]	MATCH
E5	dilate50 – ERM @100%	-0.138	-0.138 [-0.200, -0.082]	MATCH
E6	$ \Delta p $ erm @100%	+0.235	+0.235	MATCH
E6	$ \Delta p $ mask @100%	+0.490	+0.491	MATCH
E6	$ \Delta p $ mask @0%	+0.000	+0.000	MATCH
E6	$ \Delta p $ erm @0%	+0.128	+0.128	MATCH
E6	$ \Delta p $ erm @50%	+0.195	+0.195	MATCH
E7	mask – ERM, large lesions @0%	+0.249	+0.249	MATCH
E7	mask – ERM, medium lesions @0%	+0.214	+0.215	MATCH
E7	mask – ERM, small lesions @0%	+0.150	+0.150	MATCH
E7	mask – ERM, large lesions @50%	+0.087	+0.087	MATCH
E7	mask – ERM, medium lesions @50%	-0.129	-0.129	MATCH
E7	mask – ERM, small lesions @50%	-0.325	-0.326	MATCH
E7	mask – ERM, large lesions @100%	+0.064	+0.063	MATCH
E7	mask – ERM, medium lesions @100%	-0.158	-0.158	MATCH
E7	mask – ERM, small lesions @100%	-0.364	-0.364	MATCH
E14	in-lesion hair median area ratio	+0.033	+0.033	MATCH
E14	Trap A mask – ERM, area-ratio tertile T1	-0.064	-0.057 [-0.095, -0.022]	MATCH
E14	Trap A mask – ERM, area-ratio tertile T2	-0.057	-0.046 [-0.082, -0.013]	MATCH
E14	Trap A mask – ERM, area-ratio tertile T3	-0.059	-0.037 [-0.071, -0.005]	SIGN+CI
E10	hair decodability test AUROC	+0.943	+0.943 [+0.919, +0.964]	MATCH
E10	detector mean IoU	+0.220	+0.222	MATCH
E10	detector median IoU	+0.180	+0.184	MATCH
E10	LEACE A-decoder after erasure	+0.500	+0.500	MATCH
E10	clean melanoma AUROC after erasure	+0.814	+0.815	MATCH
E10	melanoma-head $ \Delta p $ after erasure	+0.058	+0.057	MATCH
E10	LEACE projection rank	+1.000	+1.000	MATCH
E12	overlap-contrast trap mask – ERM (spec: null)	+0.047	-0.103 [-0.121, -0.085]	MISMATCH
E8/E9	inpaint – ERM @50%	+0.211	+0.211 [+0.175, +0.249]	MATCH
E8/E9	inpaint – ERM @100%	+0.244	+0.243 [+0.205, +0.284]	MATCH
E8/E9	inpaint_consistency – ERM @50%	+0.046	+0.044 [-0.007, +0.092]	MATCH
E8/E9	inpaint_consistency_lam0 – ERM @50%	+0.036	+0.035 [-0.015, +0.086]	MATCH
E8/E9	leace – ERM @100%	+0.294	+0.294 [+0.249, +0.339]	MATCH
E8/E9	balanced – ERM @100%	+0.281	+0.281 [+0.236, +0.329]	MATCH
E8/E9	dfr – ERM @100%	+0.279	+0.279 [+0.220, +0.339]	MATCH
E8/E9	groupdro – ERM @100%	+0.197	+0.197 [+0.155, +0.240]	MATCH
E13	trapA mask – ERM (rev)	-0.064	-0.049 [-0.083, -0.018]	MATCH
E13	trapB mask – ERM (rev)	+0.124	+0.098 [+0.060, +0.136]	SIGN+CI
E13	trapA inpaint – ERM (rev)	+0.101	+0.098 [+0.084, +0.112]	MATCH
E13	trapB inpaint – ERM (rev)	+0.108	+0.100 [+0.083, +0.117]	MATCH
E13	trapA balanced – ERM (rev)	+0.213	+0.213 [+0.183, +0.243]	MATCH
E13	trapB balanced – ERM (rev)	+0.234	+0.204 [+0.166, +0.245]	SIGN+CI
E13	trapA dfr – ERM (rev)	+0.202	+0.196 [+0.157, +0.235]	MATCH
E13	trapB dfr – ERM (rev)	+0.192	+0.196 [+0.147, +0.251]	MATCH
E13	trapA leace_paired – ERM (rev)	+0.058	+0.074 [+0.062, +0.086]	MATCH
E13	trapB leace_paired – ERM (rev)	+0.078	+0.078 [+0.059, +0.097]	MATCH
E13	trapA leace_unpaired – ERM (rev)	+0.316	+0.320 [+0.280, +0.361]	MATCH
E13	trapA mask – ERM, HAM only	-0.030	-0.011 [-0.064, +0.043]	MISMATCH
E13	trapA mask – ERM, BCN only	-0.078	-0.067 [-0.110, -0.025]	MATCH
E15	trapA mask – ERM (rev)	-0.069	-0.104 [-0.143, -0.067]	SIGN+CI
E15	trapA balanced – ERM (rev)	+0.126	+0.146 [+0.122, +0.170]	MATCH
E15	trapA dfr – ERM (rev)	+0.114	+0.116 [+0.090, +0.145]	MATCH
E15	trapA leace_paired – ERM (rev)	+0.031	+0.034 [+0.026, +0.042]	MATCH
E15	trapA leace_unpaired – ERM (rev)	+0.192	+0.203 [+0.165, +0.243]	MATCH

7.2 The controlled sweep (E1–E3, E8–E9)

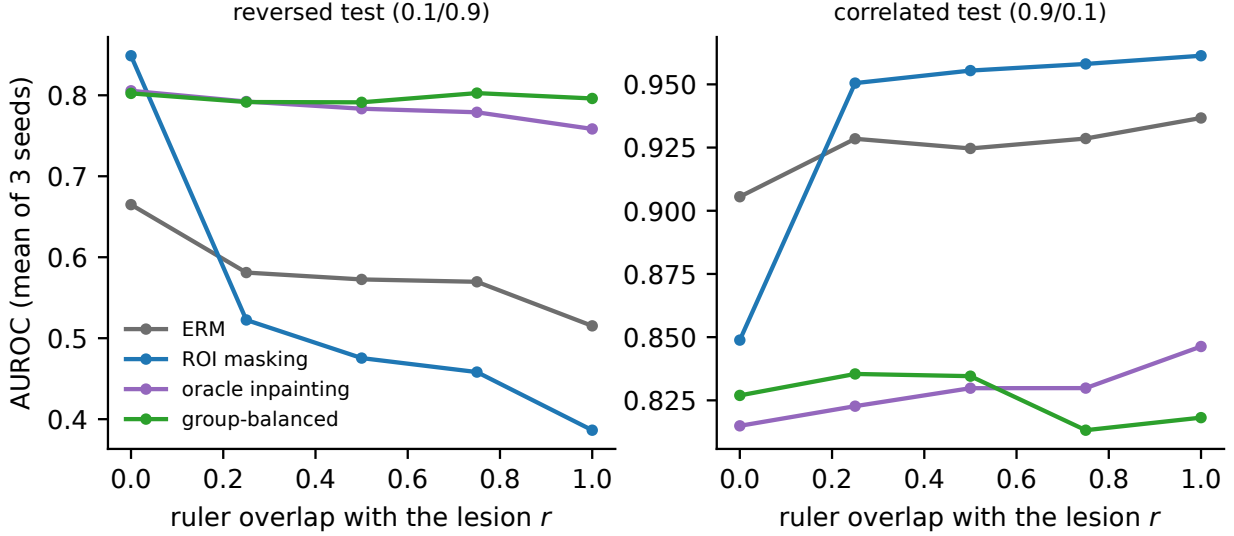


Figure 5: The pilot’s central result, re-derived: reversed-test (left) and correlated-test (right) AUROC of four arms as the ruler moves into the lesion (ISIC 2018, DINOv2 @518, mean of 3 seeds). Masking helps most when the ruler lies outside the lesion and falls below ERM once it overlaps it; its correlated-test AUROC rises with overlap — the masked model relies on the ruler it can still see.

Masking changes the reversed-test AUROC by $+0.184$ [$+0.117, +0.250$] when the ruler lies outside the lesion and by -0.129 [$-0.195, -0.064$] when it lies fully inside; the location interaction is $+0.313$ [$+0.258, +0.370$] (Figure 5). Two numbers show what happens. At 0% overlap the masked model’s correlated and reversed AUROC coincide (0.849 and 0.849): the ruler has been removed and nothing of the shortcut remains. At 100% overlap the masked model’s correlated AUROC (0.961) is *higher* than ERM’s (0.937) while its reversed AUROC collapses to 0.386 (ERM 0.515): after masking, the model uses the ruler *more*.

The reversal is not a property of one encoder or one rendering. At 224px DINOv2 gives $+0.361$ [$+0.275, +0.444$] and -0.095 [$-0.156, -0.035$] (interaction $+0.456$ [$+0.390, +0.525$]); the dermatology model DermLIP gives $+0.161$ [$+0.102, +0.222$] and -0.263 [$-0.348, -0.180$] ($+0.424$ [$+0.335, +0.517$]); a ruler whose colour, opacity, tick pattern and scale are randomised per image gives an interaction of $+0.302$ [$+0.248, +0.357$].

Methods that target the artifact rather than the background all help at full overlap: oracle inpainting $+0.243$ [$+0.205, +0.284$], group balancing $+0.281$ [$+0.236, +0.329$], DFR $+0.279$ [$+0.220, +0.339$], LEACE $+0.294$ [$+0.249, +0.339$] and GroupDRO $+0.197$ [$+0.155, +0.240$]. A training-time repair that asks the model to give the same prediction with and without the ruler (“consistency”, E8) gives $+0.044$ [$-0.007, +0.092$] at 50% overlap, and $+0.035$ [$-0.015, +0.086$] without its penalty ($\lambda = 0$): neither is significant, and the penalty adds nothing to the augmentation.

The pilot’s mechanism experiments (E4–E7) replicate numerically (Table 7): with the ruler present but uninformative, masking does not harm (E4); dilating the mask so that more ruler pixels survive increases the harm at 50% overlap (E5); the masked model’s prediction reacts about twice as strongly to inserting the ruler as ERM’s at full overlap (E6); and the harm is largest for small lesions (E7). What these observations imply about the mechanism is not interpreted in this report; only their reproducibility is.

7.3 Real hair (E10, E12–E15)

Cohort Under the specification’s rules the matched cells are: hair-free group 706 benign / 157 melanoma (pilot 708 / 157); Trap A hair-bearing 2,655 / 942 (pilot 2,903 / 1,144); Trap B 4,412 / 969 (pilot 4,093 / 809); 172 reversed-test melanomas per trap and seed, as in the pilot. The hair-free cells reproduce the pilot’s counts; the hair-bearing cells differ by 5–15% because the lesion masks of non-HAM images come from a re-trained U-Net, which moves some images across the overlap thresholds (a swap of mask source changes the overlap stratum of 6.9% of hair-bearing images and moves none from Trap A to Trap B).

E13 (DINOv2@518) Table 8. Masking *harms* when the hair lies on the lesion (-0.049 [$-0.083, -0.018$]) and helps when it lies beside it ($+0.098$ [$+0.060, +0.136$]); the crossover is $+0.147$ [$+0.111, +0.185$]. The masked Trap-A model still separates the correlated test (0.917) far better than the reversed one (0.460): it uses the hair it can still see. Every artifact-targeting arm helps in both traps; group balancing and DFR, which need only image-level hair labels, help most (C4). Paired LEACE recovers about a third of the balanced gain (C5). Unpaired LEACE “helps” by *flipping* the shortcut — its correlated AUROC (0.624) falls far below its reversed one (0.830) — which is why the pilot rejected it as unsound. The harm is concentrated in BCN images (-0.067 [$-0.110, -0.025$]; HAM -0.011 [$-0.064, +0.043$]). Masking also costs clean-test AUROC in Trap A (-0.023 [$-0.045, -0.001$]).

Table 8: Real hair traps (DINOv2@518; E13): AUROC in the three test environments (mean over 5 seeds) and change in reversed-test AUROC versus ERM (crossed 95% CI). Trap A: hair on the lesion; Trap B: hair beside it; both share the hair-free group.

Arm	Trap A: clean / corr / rev	Trap A: Δ rev vs ERM	Trap B: clean / corr / rev	Trap B: Δ rev vs ERM
ERM	0.734 / 0.920 / 0.510	—	0.716 / 0.894 / 0.510	—
ROI masking	0.712 / 0.917 / 0.460	-0.049 [$-0.083, -0.018$]	0.701 / 0.795 / 0.608	$+0.098$ [$+0.060, +0.136$]
oracle inpainting of hair	0.766 / 0.906 / 0.607	$+0.098$ [$+0.084, +0.112$]	0.738 / 0.865 / 0.610	$+0.100$ [$+0.083, +0.117$]
group-balanced	0.785 / 0.850 / 0.722	$+0.213$ [$+0.183, +0.243$]	0.752 / 0.800 / 0.714	$+0.204$ [$+0.166, +0.245$]
DFR	0.715 / 0.740 / 0.705	$+0.196$ [$+0.157, +0.235$]	0.683 / 0.669 / 0.706	$+0.196$ [$+0.147, +0.251$]
paired LEACE	0.762 / 0.913 / 0.584	$+0.074$ [$+0.062, +0.086$]	0.737 / 0.876 / 0.588	$+0.078$ [$+0.059, +0.097$]
unpaired LEACE (image-level A)	0.726 / 0.624 / 0.830	$+0.320$ [$+0.280, +0.361$]	<i>n/a</i>	<i>n/a</i>

E15 (DermLIP@224) Table 9. Masking harms in Trap A (-0.104 [$-0.143, -0.067$]) as the pilot reported, but — a finding the pilot did not test — it harms in Trap B as well (-0.090 [$-0.135, -0.046$]), so DermLIP shows no crossover ($+0.014$ [$-0.037, +0.063$]). With DermLIP, removing the background costs disease signal wherever the hair lies (clean AUROC -0.066 [$-0.089, -0.043$] in Trap A).

Table 9: Real hair traps (DermLIP@224; E13/E15): AUROC in the three test environments (mean over 5 seeds) and change in reversed-test AUROC versus ERM (crossed 95% CI). Trap A: hair on the lesion; Trap B: hair beside it; both share the hair-free group.

Arm	Trap A: clean / corr / rev	Trap A: Δ rev vs ERM	Trap B: clean / corr / rev	Trap B: Δ rev vs ERM
ERM	0.801 / 0.917 / 0.665	—	<i>n/a</i>	<i>n/a</i>
ROI masking	0.735 / 0.884 / 0.561	-0.104 [$-0.143, -0.067$]	<i>n/a</i>	<i>n/a</i>
oracle inpainting of hair	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>
group-balanced	0.836 / 0.875 / 0.811	$+0.146$ [$+0.122, +0.170$]	<i>n/a</i>	<i>n/a</i>
DFR	0.802 / 0.835 / 0.782	$+0.116$ [$+0.090, +0.145$]	<i>n/a</i>	<i>n/a</i>
paired LEACE	0.814 / 0.917 / 0.699	$+0.034$ [$+0.026, +0.042$]	<i>n/a</i>	<i>n/a</i>
unpaired LEACE (image-level A)	0.756 / 0.648 / 0.868	$+0.203$ [$+0.165, +0.243$]	<i>n/a</i>	<i>n/a</i>

E10: is hair visible to the encoder? A linear probe decodes hair presence from the DINOv2 embedding with AUROC 0.943 [0.919, 0.964] ($n = 356$ test images); in that set hair is *less* common

on melanomas (0.411) than on benign lesions (0.608). Paired LEACE on 478 images with manual hair masks (354 training pairs) needs only one direction: the hair decoder falls from 0.963 to chance (0.500), clean melanoma AUROC is unchanged (0.812 to 0.815), yet the melanoma head’s reaction to hair only falls from 0.071 to 0.057. A classical morphological hair detector (DullRazor-style) is poor: mean IoU 0.222 (median 0.184; precision 0.316, recall 0.472) — which is why the real traps use the published masks rather than a detector.

E14: how large is real hair relative to the lesion? In Trap A the in-lesion hair covers a median 0.033 of the lesion area, similar to the synthetic ruler. Masking harms in every tertile of this ratio (-0.057 [$-0.095, -0.022$], -0.046 [$-0.082, -0.013$], -0.037 [$-0.071, -0.005$]); the pilot’s “same harm at every ratio” pattern replicates, with the smallest harm in the tertile with the most hair.

7.4 E12: the overlap-contrast design

E12 labelled hair on the lesion as $A = 1$ and hair beside it as $A = 0$, with no hair-free group, and the pilot reported a null result. The re-implementation finds masking clearly harmful (-0.103 [$-0.121, -0.085$]), while balancing ($+0.160$ [$+0.144, +0.177$]) and DFR ($+0.179$ [$+0.148, +0.211$]) help. This is what the design predicts whatever the truth, because in E12 location *is* the label proxy: masking removes exactly the out-of-lesion hair that defines one side of the contrast and keeps the in-lesion hair that defines the other. The design cannot measure a location effect; E13’s two traps with a shared hair-free group can. The pilot’s null E12 result is not supported, and the design is not used further.

7.5 Audit of the pilot’s written claims

Besides the numbers, the pilot’s report made statements that were checked against the archived material (`docs/THESIS_CLAIMS_AUDIT.md`):

- **“Leakage inflated the shortcut gap by 27%”** — **not supported** as stated. The pre-correction pilot used the identical split; the 27% compared runs at different resolutions. The controlled experiment of Section 5.1 supports the direction with a magnitude of 10–18%.
- **Archived numbers** — verified: all 21 bridge intervals of the controlled sweep were recomputed from the archived predictions to 10^{-16} , and a re-run from raw images reproduced 252 per-seed AUROCs within 0.0005 (a GPU precision effect).
- **Lesion U-Net** — the pilot’s U-Net code was missing and was re-implemented. The specification’s architecture, used for the hair traps, reaches a held-out Dice of 0.89 (pilot 0.88); a ResNet-34 U-Net used for exploratory checks reaches 0.94.
- **Mask-source sensitivity** — the pilot reported that about 11% of images change overlap stratum when the mask source is swapped; the re-implementation finds 6.9%, and no image moves between Trap A and Trap B.
- **E11 (artifact atlas and vignetting trap)** was not re-implemented: ink appears on the lesion in too few images (the traps exclude ink), and vignetting is a frame artifact that lies outside the lesion by construction, so it cannot inform the in-ROI question.

7.6 The mismatches

1. **E1, masking at 25% overlap.** The pilot’s interval excluded zero; the crossed interval includes it. The harm is significant from 50% overlap onward.

2. **E4, occlusion control at 0% and 100% overlap.** The pilot reported small positive effects of masking with the ruler present but uninformative; with the crossed estimator they are not significant. The conclusion — occlusion by the mask does not harm — is unchanged.
3. **E12.** Not null (above); a structural problem of the design.
4. **E13, HAM-only stratum.** The pilot reported harm in HAM images; the replication does not ($-0.011 [-0.064, +0.043]$). The harm is carried by BCN images.

None of these affects the pilot’s central claim — that masking reverses from helpful to harmful as the ruler moves into the lesion — which replicates with every encoder and ruler rendering.

8 What failed or changed in this stage

- The pilot’s interval estimator was too narrow whenever seeds share test images; it was replaced, and three pilot significance claims (E1 at 25%, E4 at 0% and 100%) no longer hold.
- E12’s null result and the “27%” leakage figure are not supported; the HAM-only harm of E13 does not replicate.
- DermLIP shows no hair crossover in the replication: it is harmed by masking at both hair locations.
- The hair-bearing cells of E13 differ by 5–15% from the pilot’s because the lesion U-Net had to be re-trained.

9 Anticipated questions

Why re-implement instead of re-running the pilot’s code? Most of the pilot’s code for E9–E15 was not in the archive, and re-running code verifies only that it runs. Re-implementing from a written specification tests whether the *description* is sufficient to reproduce the result, which is what a reader of a paper can do.

Why is the reversed test the primary metric, not the clean test? The clean test measures disease recognition without the shortcut, so it cannot show whether a model *uses* the artifact. The reversed test penalises reliance on the artifact directly. Both are always reported, together with the correlated test, so that a method that simply inverts the shortcut (as unpaired LEACE does) is visible.

Why 0.9/0.1? It is strong enough that every encoder learns the shortcut, which is the regime where mitigation matters, and weak enough that both classes contain artifact-free and artifact-bearing images in training. The same values are used everywhere, so effects are comparable.

Why frozen encoders and linear heads? Every arm then uses identical features; differences between arms are caused by the input (masked or not) or the head, not by optimisation noise. It is also the common way foundation models are deployed on small medical datasets.

Why choose C and the threshold on clean validation data? Choosing them on data with the shortcut would reward the shortcut; choosing them on reversed data would leak the test condition into training.

Why group by perceptual hash and not by patient? ISIC 2018 does not publish patient identifiers; lesion identifiers and near-duplicate detection are the strongest grouping available. The leakage experiment shows that this grouping matters (10–18% gap inflation otherwise).

Why a crossed bootstrap and not simply more seeds? More seeds reduce training variance but do not create new test images. The quantity of interest is a difference on a finite test set; its uncertainty must include the sampling of that set once, which is what the shared image weights do.

Why do the E13 cell counts differ from the pilot’s? The hair-free group reproduces exactly. The hair-bearing cells depend on the lesion masks of non-HAM images, which come from a re-trained U-Net; images near the 0.1 and 0.5 overlap thresholds move between strata. No image moves between Trap A and Trap B.

Why does unpaired LEACE look best, and why is it rejected? It removes the direction that separates images with and without hair *in the training data*, where hair is strongly associated with the label; that direction contains much of the diagnosis, and the model ends up predicting the opposite of the shortcut (correlated AUROC far below reversed). Paired LEACE compares the same image with and without hair and removes only the hair.

Is 2,437 images enough? For the controlled sweep, yes: the location interaction is large relative to its interval (+0.313 [+0.258, +0.370]). Individual overlap cells are less precise, which is why the 25% cell lost significance.

10 Conclusion

Supported. The pilot’s central result replicates from its written specification: masking helps when the ruler lies outside the lesion and harms when it overlaps it (interaction +0.313 [+0.258, +0.370]), with two encoders, two resolutions and a randomised ruler; with real hair on DINOv2 masking harms in-lesion and helps out-of-lesion (crossover +0.147 [+0.111, +0.185]); methods that need only an image-level artifact label fix both traps; paired linear erasure is insufficient. **Not supported.** The pilot’s null E12 result; the “27%” leakage figure; significance at 25% overlap and in the occlusion control; harm in HAM images; a hair crossover with DermLIP. **Fixed for the rest of the work.** Grouped splits, clean-validation model selection, pre-registration, and the crossed bootstrap.

11 Questions for the supervisor

1. Is the reversed-environment AUROC acceptable as the primary metric for the thesis, or should a clinical metric (sensitivity at a fixed specificity) be primary from the start?
2. How should the thesis present the estimator correction: as a methods detail, or as a contribution in its own right (it affects any study that averages several training seeds on one test set)?
3. Should the E12 design and its failure be reported in the thesis (as a warning about contrast designs) or only in the appendix?
4. Is it acceptable that the pilot’s registration is its written specification, committed after the pilot ran?

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